

Kourosh Salahi

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EDUCATION

UC Berkeley - Management, Entrepreneurship, & Technology (M.E.T.) Program Graduating 2027

Double Major: B.S. Electrical Engineering & Computer Science (EECS), B.S. Business Administration GPA: 3.87

Distinctions: Selected as one of ~50 students to be part of the M.E.T. Program (<2% acceptance rate); Dean's List (x4)

Relevant Coursework: Reinforcement Learning, Machine Learning, Deep Neural Networks, Operating Systems, Computer Vision, Robotics, Efficient Algorithms, Probability Theory, Signal Processing, Financial Accounting, Business Analytics

PROFESSIONAL EXPERIENCE

- **DoorDash Inc.**, *Software Engineering Intern* *May 2026 - August 2026*
 - Built a policy impact simulator to replay order resolutions through new refund policies prior to rollout.
 - Delivered pre-deployment estimate of financial exposure for credits and refunds, and cohort fairness.
 - Integrated LLM-based evaluation layer enabling data-driven policy vetting on customer fairness and dispute risk.
 - Finished main project a month early, then implemented an order-value-based credit cap, saving over **\$300k** per year.
- **Google Inc.**, *Software Engineering Contractor* *September 2025 - February 2026*
 - Developed CLI testing tool to capture and replay PostgreSQL traffic for performance benchmarking.
 - Implemented workload capture, concurrency preservation, and variable-speed replay modules in Python.
 - Automated summary report generation after replays to benchmark transactions per second and query latency.
- **GoodFin Inc. (YC W22)**, *Software Engineering Intern* *May 2024 - August 2024*
 - Created an administrator dashboard with AWS (DynamoDB, Amplify, Lambda), React, and Node to manage user investments, company metrics, and products; backed by serverless REST APIs and S3-based file handling.
 - Reduced engineering intervention time and shortened investment management tasks from **hours to seconds**.
 - Built SMS marketing subscription flow with Klaviyo and DynamoDB, increasing customer engagement **by 27%**.
 - Built automated premium-user welcome emails with Stripe webhooks and IAM data, boosting retention **by 11%**.

RESEARCH AND TEACHING

- **Berkeley AI Research, Prof. Trevor Darrell Group**, *Research Assistant* *July 2025 - September 2025*
 - Conducted research on diffusion-based generative computer vision models for 6-DOF robotic perception and control.
 - Optimized diffusion training pipeline by replacing Diffusion Transformer (DiT) loss with Representation Alignment (REPA) and Scalable Interpolant Transformer (SiT) loss, achieving a **30%** performance improvement.
- **UC Berkeley EECS Department**, *Undergraduate Course Staff* *August 2025 - Present*
 - Head Lab TA for graduate level robotics class. Led team of 10 Lab Staff in developing and administering labs.
 - Served as course staff for Berkeley flagship Machine Learning course, CS189; enhanced learning for **800+ students**.

PROJECTS AND EXTRACURRICULARS

- **VR Teleoperated Robotic Arm:** Developed a teleoperation system for a UR7e arm using ROS2 and OpenCV, enabling real-time control through a Meta Quest VR headset. Implemented color and shape based object detection and created a motion-recording pipeline allowing different captured trajectories to be replayed autonomously based on workspace objects.
- **RL Trained Robotic Hand:** Designed and trained PPO-based RL policies for dexterous manipulation in Isaac Lab/Isaac Sim, implementing fingertip-distance observations and spin-penalty reward shaping; deployed policies to a physical Allegro hand via ROS2 with EMA-smoothed joint control and keypoint-based cube pose estimation
- **Autonomous Drone Racing via NLMPC:** Engineered a two-phase nonlinear MPC system in NVIDIA Isaac Sim, achieving a 16.66s lap time and 96% success rate across 50 trials on a 12-gate course. Built an offline minimum-snap trajectory planner and real-time 20-step NLMPC solver using CasADi for 100Hz control.
- **Neural Radiance Field:** Built a NeRF from scratch using an 8-layer MLP with sinusoidal positional encoding and a skip connection, feeding separate density and view-dependent color heads. Achieved 23.9 dB PSNR on held-out views and produced a 3D reconstruction of my own captured object from novel angles.

ADDITIONAL INFORMATION

- **Programming Languages:** Python, Java, C++, C, SQL, Go, JavaScript, HTML, CSS.
- **Technologies:** PyTorch, Numpy, Pandas, MuJoCo, OpenCV, ROS2, SLAM, Isaac Sim, React, AWS, PostgreSQL.
- **Leadership:** Eta Kappa Nu Honor Society Industry Relations Officer, Undergraduate Real Estate Club Analyst.
- **Languages:** Trilingual - Fluent in English and Farsi, proficient in Spanish.
- **Interests:** Second Degree Black Belt in TaeKwonDo, Judo Beginner, Golfer, Guitarist, Fragrance Collector.